A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 28

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Asbestos:** fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma – disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) **2 Cyanide:** very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested – disposal: incineration so carbon and nitrogen are separated and oxidised to CO2 and NOx (much smaller problem) **2 High-level radioactive waste:** intense ionising radiation, heat generation, long half-lives – disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **vitrification** → sealing waste into glass **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6

Do not write outside the box

Question	Marking guidance	Marks
01.2	1 mark for the point below. **1** The fibres are bound / cannot become airborne so cannot be inhaled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material **1** The solid cement immobilises the waste so it cannot leak / leach out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
01.4	1 mark for the point below. **1** Even if the glass shatters the waste stays encapsulated in the fragments – glass is more stable / resistant to heat and leaching over very long timescales **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Water = 65.0 - 55.0 = 10.0 g – fresh soil = 40.0 g **1** 10.0 / 40.0 x 100 = 25.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Organic matter = 55.0 - 52.6 = 2.4 g – dry soil = 30.0 g **1** 2.4 / 30.0 x 100 = 8.0% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** So all water is removed and the mass loss on burning is due to organic matter only **1** Heating until mass no longer changes shows all water has evaporated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** Organic matter releases nutrients as it decomposes and provides food for soil biota **1** Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration **1** Surface organic matter / humus adhesion reduce rain splash and wind erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	3
02.5	1 mark for the point below. **1** Samples taken from different depths / locations (organic matter is concentrated near the surface) – incomplete burning – different heating times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Stress-related illness:** ulcers, high blood pressure, heart disease **1 Behavioural changes:** irritability, aggression, lack of concentration leading to accidents **1** Communication problems / educational underperformance (eg schools under flight paths) **1** Sleep disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1 Acoustic fatigue:** objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) **1 Vibration damage:** repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes **1 Shock impacts:** the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Birds that sing to establish territories / attract mates cannot be heard against background noise **1** Birds scared by sudden noise leave nests so predators take eggs / chicks – cliff-nesting seabirds knock eggs or chicks into the sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Whales / dolphins use hearing / echolocation to find food and communicate – loud sonar causes hearing damage if close to the source **1** Behavioural changes / disorientation possibly linked to mass strandings – reduced feeding success **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Discharge of tank-washing water from tankers **1** Oil rig / drilling accidents / leakage during drilling **1** Pipeline leaks **1** Leakage from storage tanks / refinery spills **1** Waste lubricating oil from vehicles / industry washed into drains **1** Other ship accidents (fuel oil) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia **1** Feathers lose buoyancy / waterproofing so heavily oiled birds drown **1** Birds swallow toxic oil when preening – time spent cleaning feathers reduces time for feeding themselves / their young **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen **1** Toxic components (benzene, xylene) dissolve and poison plankton near the surface **1** Oil blocks light so algae / phytoplankton cannot photosynthesise **1** Oil masks odours / narcotises so animals cannot find mates or food by smell **1** Shore organisms (molluscs, crustaceans) are covered and asphyxiated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Dispersal / dilution by currents and waves is prevented so oil stays concentrated – in open water most spills disperse in days or weeks **1** Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) **1** Radon travels up cracks in the rock and through foundations into buildings **1** It becomes concentrated inside enclosed / poorly ventilated buildings – control by ventilating foundations / sumps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16 + 1.5 = 17.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

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Question	Marking guidance	Marks
05.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Where they live (geology / radon, altitude) **1** Occupation (radiographer, nuclear worker, aircrew, miner) **1** Medical history (number of X-rays / CT scans) **1** Lifestyle / diet (eg shellfish near discharges) – smoking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
05.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it – risks can be reduced (shielding, dose limits) **1** Symptoms of chronic low-dose exposure take a long time to develop **1** The same symptoms (cancers) have other causes so effects are hard to attribute **1** Accurate data on human exposure / doses is often unavailable (especially after accidents) **1** People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **Accurate** → close to the true value	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** A: Student's t-test - a difference between the means of two sets of normally distributed data **1** B: chi-squared test - a comparison of observed frequencies (counts) with expected frequencies **1** C: Spearman's rank correlation - a relationship between two variables measured on the same sampling points **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Student's t** → a test for a difference between the means of two normally distributed sets • **chi-squared** → a test comparing observed frequencies with expected frequencies • **Spearman** → a test for correlation between two sets of ranked data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 2% probability, or a 1 in 50 chance, that a result at least as extreme as this would arise by chance alone if the null hypothesis were true **1** This is below the 5% significance level normally used, so the null hypothesis is rejected and the difference or correlation is described as statistically significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **significance level** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** The calculated value on its own has no meaning: it must be compared with the value that would be exceeded by chance in 5% of cases, which is read from a table of critical values **1** The critical value depends on the number of degrees of freedom, which comes from the number of categories or pairs of data, because a larger sample makes a given result less likely to be chance **1** If the calculated value exceeds the critical value at p = 0.05 the null hypothesis is rejected – if it does not, the null hypothesis is not rejected and the result is not significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **critical value** → the table value the calculated statistic must reach to be significant • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3

Do not write outside the box

Question	Marking guidance	Marks
08.4	**1 mark each** 1 mark for each of the 3 points below. **1** The standard deviation measures the spread of the individual values about the mean, so a small value means the heights were closely grouped and the mean is a good summary of them **1** Comparing the standard deviations with the difference between the means shows whether the two sets really are distinct or merely overlap **1** It is drawn as error bars above and below each plotted mean – where the error bars of two means do not overlap, a significant difference is likely, and where they overlap widely it is not **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	3
08.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** A correlation shows only that the two variables change together – it suggests a link that further work would have to test **1** A third factor may cause both, for instance shade, which raises soil moisture by reducing evaporation and also directly favours the moss **1** The relationship may run the other way round: a dense moss cover holds water and shades the soil, so the moss could be raising the moisture rather than responding to it **1** The points along one transect are close together and not independent, so a pattern may reflect one gradient across the site rather than a general relationship **1** Only an experiment in which soil moisture is changed while every other variable is held constant can establish cause **1** A result significant at p = 0.05 will still arise by chance about 1 time in 20, so a single significant result is not proof **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together • **transect** → a line along which samples are taken to show change with distance • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407, 420-423; AQA spec 3.7.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well – pump-jacks ('nodding donkeys') increase flow **1** Pressure falls as oil is removed so the remaining oil no longer flows to the well **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) **1** Captured CO_2 can be pumped down instead, increasing oil recovery while storing the CO_2 underground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground	2
09.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity **1** Controlled underground combustion to heat the oil **1** Detergents / solvents to reduce surface tension so oil flows more easily **1** Bacteria that partially digest heavy oil into lighter oils and produce CO_2 to maintain pressure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.4	**1 mark each** 1 mark for each of the 3 points below. **1** Many wells can be drilled from a single platform, reducing land take / number of rigs **1** Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached **1** Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable – demand fluctuates daily, weekly, seasonally and short-term – supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants – large-scale electricity storage not yet possible directly so energy must be converted • **Pumped-storage HEP:** proven, rapid response, peak shaving, but limited sites • **Batteries:** lithium-ion for grid and vehicles – factors affecting viability – V2G • **Hydrogen:** electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid – high energy density enables vehicle use • Compressed air with heat recovery – thermal storage (molten salt for CSP, high-volume stores, thermal mass) – flywheels – supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage – energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **HEP** → electricity generated from falling water • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (9 marks)	
	Renewable electricity is intermittent - solar stops at night, wind on calm days, tidal power at slack water - and some of it is unpredictable, while demand fluctuates by the minute, day, week and season, so supply and demand rarely coincide. Without storage, surplus renewable output is wasted and shortfalls must be met by fossil-fuel peaking plants, which defeats the purpose. Electricity itself cannot be stored at scale, so the energy must be converted into another form and back.	
	Pumped-storage HEP is proven and responds within seconds for peak shaving, but suitable sites are few. Lithium-ion batteries now work at grid scale and in vehicles, with vehicle-to-grid schemes turning car batteries into distributed storage, though lithium supplies and lifetime limit them. Hydrogen from electrolysis can be stored, used in fuel cells or injected into the gas grid (power-to-gas), and its high energy density suits vehicles, although each conversion loses energy. Compressed air with heat recovery, molten-salt thermal stores for concentrating solar plants, thermal mass in buildings, flywheels and supercapacitors each fit particular timescales.	
	Storage is not the only answer: a diverse mix of wind, solar, tidal and biofuel smooths supply, smart-grid management shifts demand, and conservation reduces peaks. But these reduce rather than remove the mismatch. Short-term storage can be batteries and pumped hydro; the harder problem is seasonal storage, where only hydrogen or fuels made from surplus electricity look plausible. Without both, renewables can supplement but not dominate a country's supply.	
	Weaker answer - Level 2 (5 marks)	
	Renewable energy is intermittent. Solar panels do not work at night and wind turbines do not work when it is not windy, but people still need electricity then. Storage means the energy can be saved for when it is needed. Pumped storage pumps water uphill when there is spare electricity and lets it flow back through turbines at peak times. Batteries can also store electricity. Hydrogen can be made from electricity and used in fuel cells. So storage is needed for renewables to provide most electricity.	
	What makes the difference	
	• Level 2 explains the mismatch and describes three storage methods correctly, but does not evaluate them (sites, lithium, conversion losses, timescales).	
	• Level 3 distinguishes short-term from seasonal storage - the key insight for 'most of a country's electricity'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 also considers the alternatives to storage (diversity, grid management, conservation) and explains why they are insufficient alone.	
10.2	1 mark for the point below. **1** Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why non-renewable:** formed too slowly – geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock – Lasky's principle – cut-off ore grade and its drivers • **Extraction technologies:** mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • **Reducing demand:** recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • **Environmental limits:** impacts of mining and their control – energy costs rising with lower grades – conflicts	25
	• **Judgement:** technology can extend supplies substantially but only within energy and environmental constraints – circular economy essential *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • **How properties predict behaviour:** dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • **Control based on properties:** persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones – systemic pesticides and seed coatings • **Heavy metals:** lead uses discontinued, storage at high pH to reduce solubility – mercury chemical form – radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays) – oil properties (floats, insoluble) exploited by booms and skimmers	25

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Question	Marking guidance	Marks
	– density in cyclone separators • **Principles:** CPA and CGM based on properties – emission location / timing – ALARA and BATNEEC – polluter pays – precautionary principle • **New technologies:** satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • **Evaluation:** prevention better than cure – limits of remediation – cost-effectiveness – contrast success of ozone (Montreal) with climate change – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	
	Model answer - Level 5 (22 marks)	
	Pollutants cause damage in proportion to how much reaches sensitive organisms and how harmful it is when it gets there. Both depend on properties, so knowing a pollutant's properties lets us predict where it will go and which control will work.	
	State of matter and density decide dispersal. Solid particulates settle near the source unless they are very small: PM10 and smaller stay suspended, travel far and penetrate lungs, so smoke is controlled at the point of release with electrostatic precipitators, cyclones and bag filters, and smokeless zones prevent its production. Gases disperse widely, and dense gases such as the hydrogen cyanide released at Bhopal hug the ground, so location and emission height matter. Persistence decides whether dilution is acceptable. Sewage and other biodegradable wastes break down, so treatment plus dilution in a river or the sea is sufficient; organochlorine pesticides, CFCs, heavy metals and long-lived radioisotopes are not broken down, so they travel, accumulate and must be contained, banned or replaced - DDT and CFCs were both controlled by substitution once their persistence was understood.	
	Solubility governs the pathway. Water-soluble nitrate leaches into rivers and aquifers, so it is controlled by nitrate vulnerable zones, fertiliser timing and buffer strips; liposoluble pollutants such as DDT and methyl mercury pass into fat and are biomagnified along food chains, as at Minamata, so even low concentrations in water are dangerous and the only effective control is to stop release. Adsorption on clay immobilises caesium and metals, which is why radioactive discharges are monitored in sediments and why activated carbon and polymer adsorbents are used deliberately in effluent treatment.	

Do not write outside the box

Question	Marking guidance	Marks
	Toxicity, specificity and mode of action set acceptable doses and identify species at risk. Neonicotinoids are toxic to insects but not vertebrates because they cannot cross the vertebrate blood-brain barrier, which explains both their popularity and their effect on bees; pyrethroids replaced organochlorines because they are less persistent. For mutagens and carcinogens such as ionising radiation there is no safe dose, so exposure is kept As Low As Reasonably Achievable with shielding, distance, time limits and monitoring. Reactivity explains secondary pollutants - sulfur dioxide oxidised to sulfuric acid, nitrogen oxides and hydrocarbons forming ozone and PANs in photochemical smog - so control targets the primary emissions with flue-gas desulfurisation and catalytic converters. Synergism between smoke and sulfur dioxide, or ozone and sulfur dioxide, means limits must consider pollutants together.	
	Properties interact with the environment: temperature inversions trap pollutants, currents disperse them, pH and oxygen alter solubility. Critical Pathway Analysis combines the two to predict where a discharge will concentrate and to focus monitoring, and Critical Group Monitoring checks the most exposed people. These principles have reduced smoke, sulfur dioxide, lead and organochlorines dramatically in developed countries. Their limits are pollutants with diffuse sources or that underpin economies - nitrate, carbon dioxide - where knowing the properties is not the problem; changing behaviour is. Even there, the precautionary principle applies the same logic: predict the properties of a new substance from similar ones and control it before release, because prevention is always cheaper than remediation.	
	What makes the difference	
	• Every paragraph is organised around a **property**, each linked to a prediction and a named control - the essay answers the question throughout rather than describing pollutants.	
	• Examples span the specification (smoke, acid rain, pesticides, heavy metals, radiation, nutrients, oil), with quantitative and named detail (Bhopal, Minamata, PM10, ALARA), which is what Level 5 breadth means.	
	• The final paragraph evaluates: where the approach has worked, where it has not and why, and reaches a justified conclusion.	
	• A Level 3 version of this essay would describe several pollutants and their effects accurately but only occasionally connect a property to a control, and would end without a judgement.	

Do not write outside the box